Machine-Learning-in-Ecology-R-Workshop

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And lastly.... RES Aquatic Insect SIG Lead

Introduction

- I am undertaking a PhD which uses machine learning, **I am no expert**, I just want to share, lets see how we get on.
- There will be a lot of questions from this session, too many to fit in the workshop but don't worry I have the solution at the end..

Worries?

- What should I use to code with? R? or Python? Both?
- Can I use this method without being a data scientist or similar?
- What if its too much to learn for a 3-hour workshop?
- Where do I go from here?

There are a lot of great videos online! I am just another facilitator!

Workshop Outcomes

- An introduction to Machine Learning in Ecology
- We will answer a scientific question
- A code along (at pace) of a script that will run machine learning
- Chance to ask questions
- You will have homework!

What is Machine Learning (ML)?

- Definition: Algorithms that learn patterns from data without explicit programming
- Contrast with traditional statistics
- Key terminology: training data, features, predictions, model

Machine Learning Explained in 100 Seconds

YouTube · Fireship · 9 Sept 2021

YouTube

Resources:

1. Breiman, L. (2001). "Statistical Modelling: The Two Cultures" - classic paper on ML vs statistics
2. James et al. "Introduction to Statistical Learning" (free PDF) - Chapter 1

ML Types – The main 2:

Other types: **reinforcement learning** like Chatgpt etc.

1) Supervised Learning

- **What**: Learning from labelled data (known outcomes)
- **When**: You have response variable + want to predict/classify
- **Examples in ecology**:
  - Predicting species richness from environmental variables (this workshop!)
  - Species distribution models
  - Classifying habitat types from remote sensing

2) Unsupervised Learning

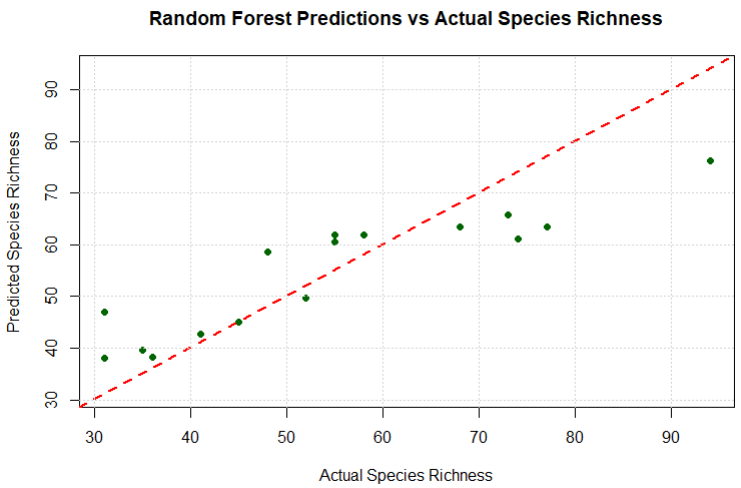
- **What**: Finding patterns in unlabelled data (no known outcomes)
- **When**: Exploring data structure, finding groups, reducing dimensions
- **Examples in ecology**:
  - Clustering similar communities (NMDS, k-means)
  - Identifying habitat types from multivariate data
  - Ordination techniques

Kick the habit…(ecologist like to use what others use)… don't just pick a package…

You don’t need to take all this in…

Supervised Methods/algorithms:

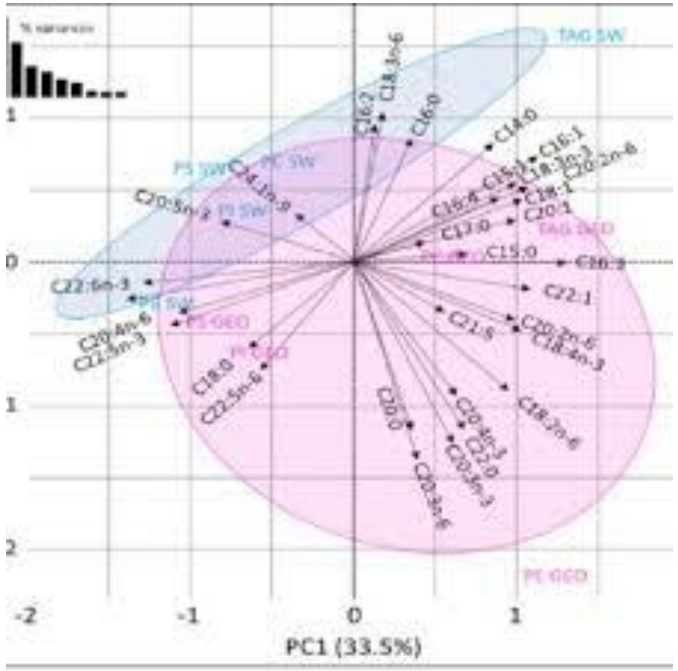
Method	Type	When to Use	Ecological Example
Random Forest	Classification/ Regression	Complex non-linear relationships	This workshop!
Logistic Regression	Classification	Binary outcomes	Species presence/absence
Linear Regression	Regression	Linear relationships	Traditional ecology
Gradient Boosting	Classification/ Regression	High accuracy needed	Competition predictions
Neural Networks	Both	Very large datasets	Image classification (camera traps)



Similarly…

Unsupervised Methods/algorithms:

Method	Purpose	When to Use	Ecological Example
K-means clustering	Group similar observations	Find community types	Site clustering by species comp
PCA	Dimensionality reduction	Many correlated variables	Environmental gradients
NMDS	Ordination	Community composition	Classic community ecology
Hierarchical clustering	Create dendrograms	Visualize relationships	Taxonomic relationships



"When Should I Use ML?"

Use Supervised ML when:

- ✅ You have a clear response variable (species richness, abundance, presence/absence)
- ✅ Relationships might be non-linear or complex
- ✅ You want to predict new/future observations
- ✅ You have enough data (>50-100 observations for Random Forest)
- ✅ Prediction accuracy matters more than simple interpretation

Use Unsupervised ML when:

- ✅ Exploring patterns without predetermined response
- ✅ Identifying natural groupings in your data
- ✅ Reducing dimensionality of complex datasets
- ✅ Hypothesis generation rather than testing

Stick with traditional statistics when:

- ✅ Relationships are likely linear (when one thing increases/decreases another increases/dec)
- ✅ Small sample sizes (<30) .... **We will come back to this later**
- ✅ Mechanistic interpretation is critical (e.g. richness = 2 x altitude + 1.5 x depth)
- ✅ Hypothesis testing rather than prediction

1: Install R

RStudio requires R 3.6.0+. Choose a version of R that matches your computer's operating system.

R is not a Posit product. By clicking on the link below to download and install R, you are leaving the Posit website. Posit disclaims any obligations and all liability with respect to R and the R website.

DOWNLOAD AND INSTALL R

2: Install RStudio

DOWNLOAD RSTUDIO DESKTOP FOR WINDOWS

Size: 314.19 MB | SHA-256: 84442594 | Version: 2026.0
Released: 2026-02-19

Hi Arron
Where should we start?

Ask Gemini 3

Tools

Fast

Create image | Create music | Boost my day | Help me learn | Write anything

RStudio

Hey, what's on your mind today?

a product | Write a first draft | Take a quiz

guage | Get advice | Paint a sunrise

Explained: Generative AI's environmental impact

Rapid development and deployment of powerful generative AI models comes with environmental consequences, including increased electricity demand and water consumption.

Adam Zewe | MIT News
January 17, 2025

```
# CHANGE THIS to match your folder structure
# setwd("~/path/to/your/workshop/folder")

# Check your working directory
getwd()
```

Console | Background Jobs

R 4.3.0 · C:/Users/axw213/Desktop/R Studio/workshop_24_02_2026/machine learning workshop/

```
> varImpPlot(rf_model,
+            main = "Variable Importance for Species Richness",
+            pch = 19,
+            col = "darkblue")
>
> # The plot shows two measures:
> # %IncMSE: How much prediction error increases when variable is excluded
> # IncNodePurity: Total decrease in node impurities from splitting on variable
>
>
> ################################################################################
> ################### PHASE 13: MODEL PREDICTIONS & EVALUATION ###################
> ################################################################################
>
```

Lets begin!... What's the question?

Space and time! – can we use both?

Other features?

Width and depth

Diversity and abundance = richness

Do river morphological features drive taxon richness over space and time?

Environment Agency data

River Habitat Survey - Survey Details and Summary Results

Published by: Environment Agency
Last updated: 10 October 2025
Topic: Not added
Licence: Other Licence
View licence information

Summary

This record is for Approval for Access product AfA434. "River Habitat Survey (RHS) is the Environment Agency standard for collecting data on the physical character and quality of river habitats across the UK. This dataset provides survey details and summary results for river habitat surveys carried out from 1994 to present. This dataset is a subset of AfA286 River Habitat Survey. Since 1994 OVER 21,000 surveys have been carried out in

View full summary

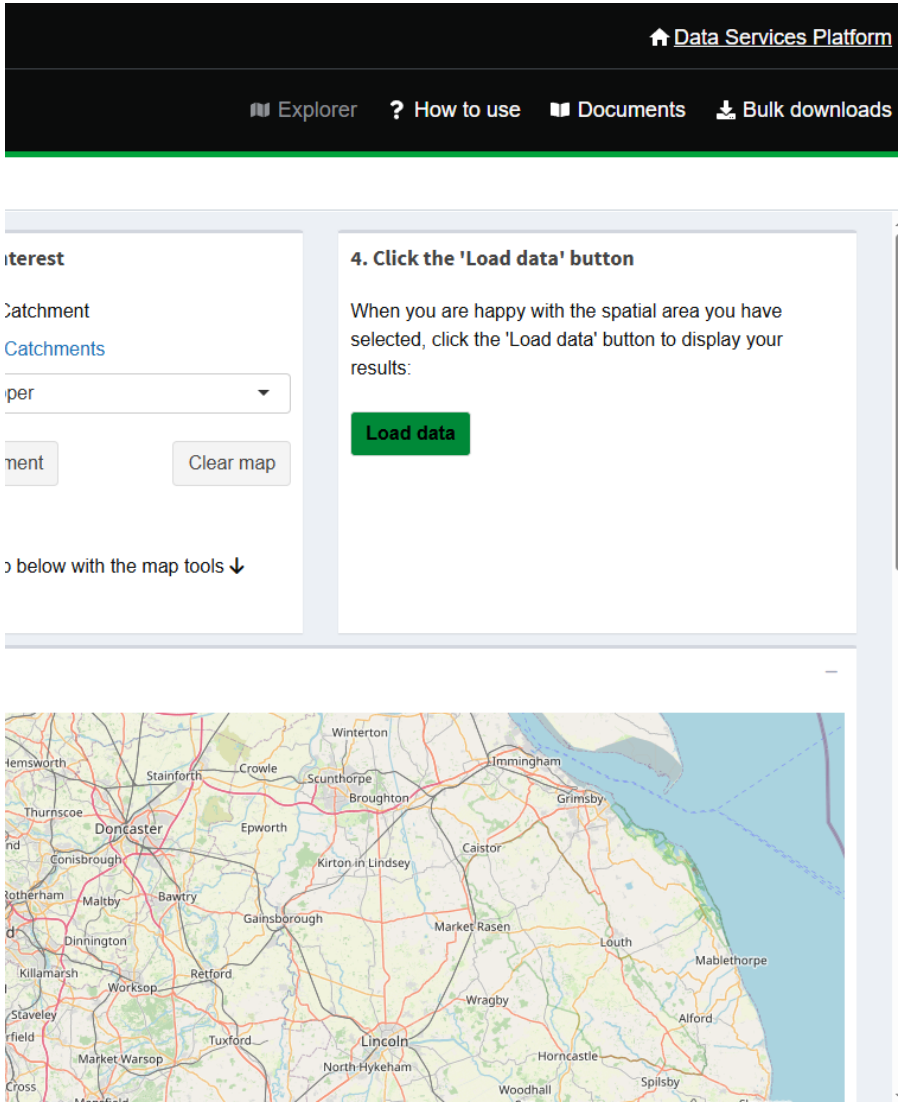
More from this publisher
All datasets from Environment Agency

Search

We will use 2020-2026 data (remember this)

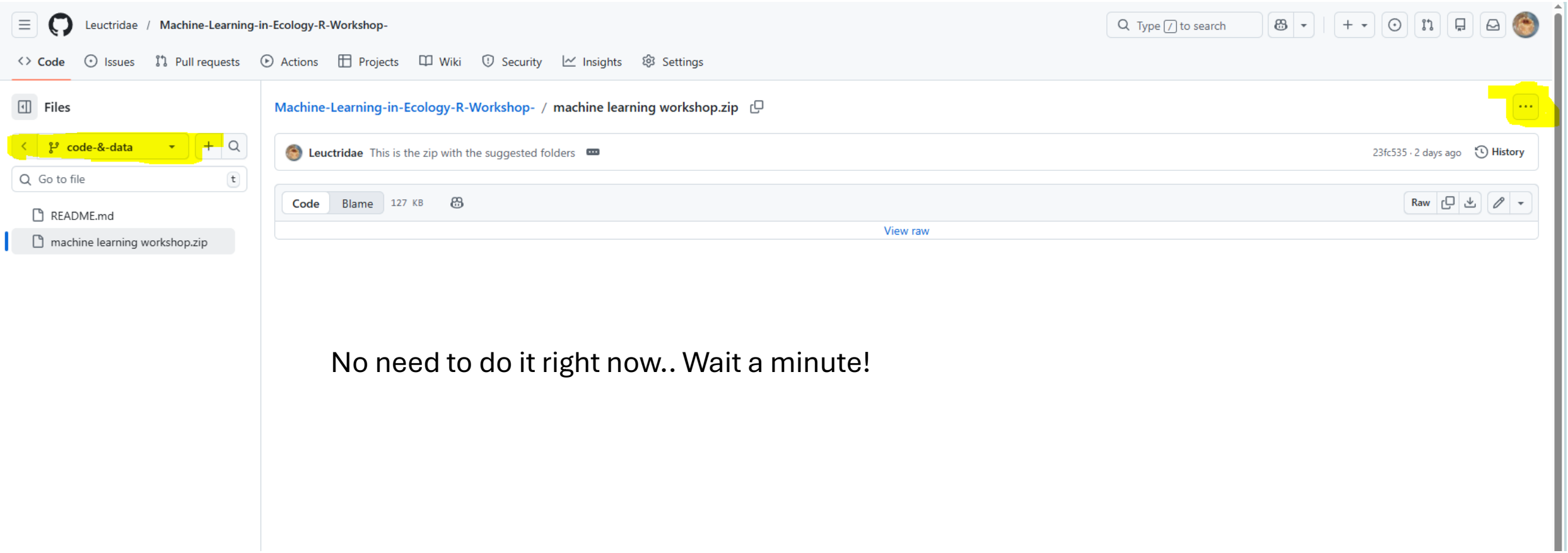
Data links

Link to the data	Format	File added	Data preview
River_Habitat_Survey_-_Survey_Details_and_Summary_Results_(2).zip	ZIP	10 October 2025	Not available



https://environment.data.gov.uk/ecology/explorer/

Download the data

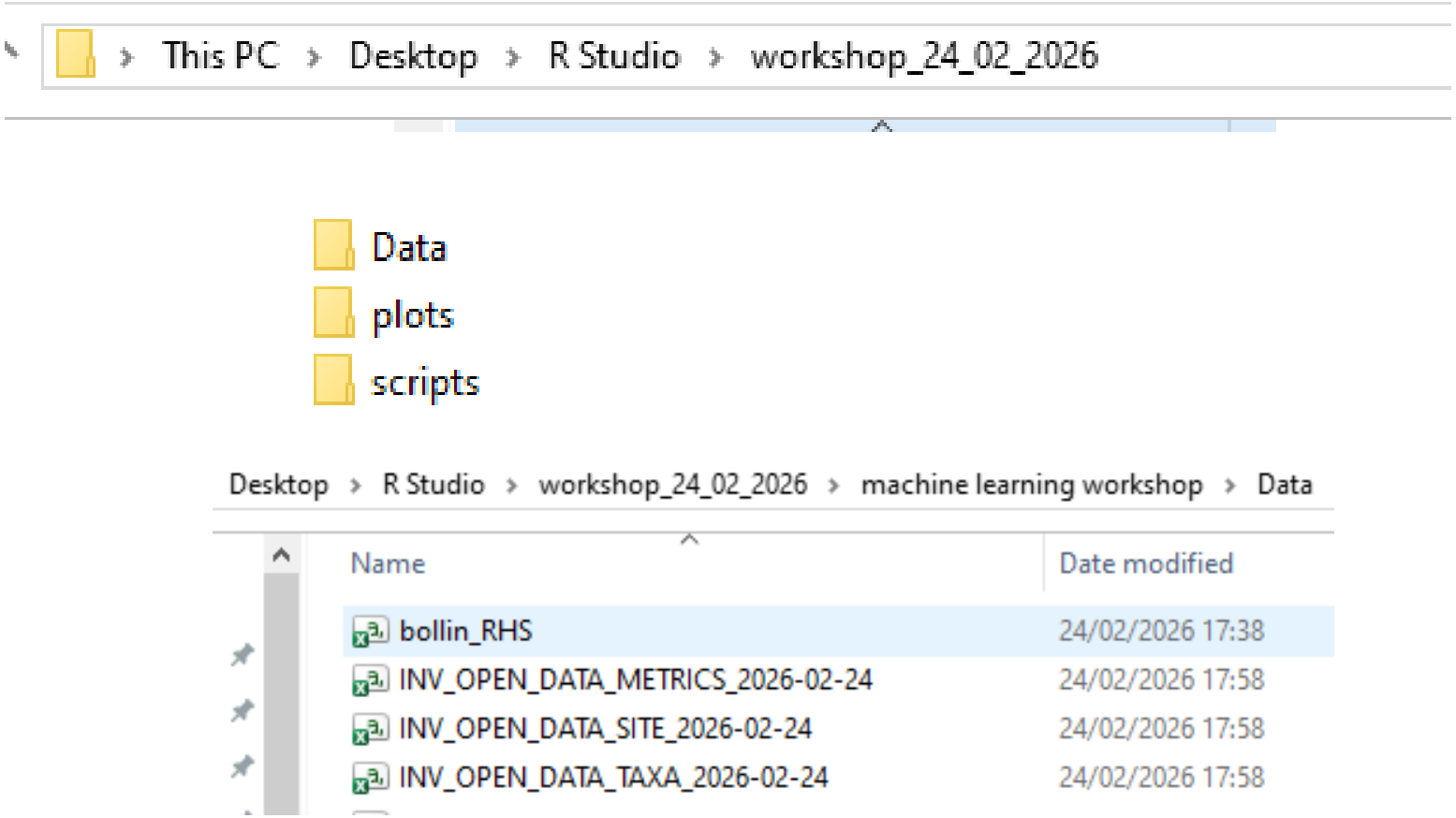


No need to do it right now.. Wait a minute!

https://github.com/Leuctridae/Machine-Learning-in-Ecology-R-Workshop-

Structure!!!

- Folder for project
- Folder for Data
- Folder for Script
- Folder for plots
- Simple…



Coding? It takes time… I forget all the time..

```
##########################################################################
######################### MACHINE LEARNING IN ECOLOGY ######################
################ R WORKSHOP: INVERTEBRATE COMMUNITY ANALYSIS ################
##########################################################################
#
```

```
# Load required libraries
library(ggplot2)       # For plotting
library(vegan)         # For ecological diversity calculations
library(randomForest)  # For Random Forest models
```

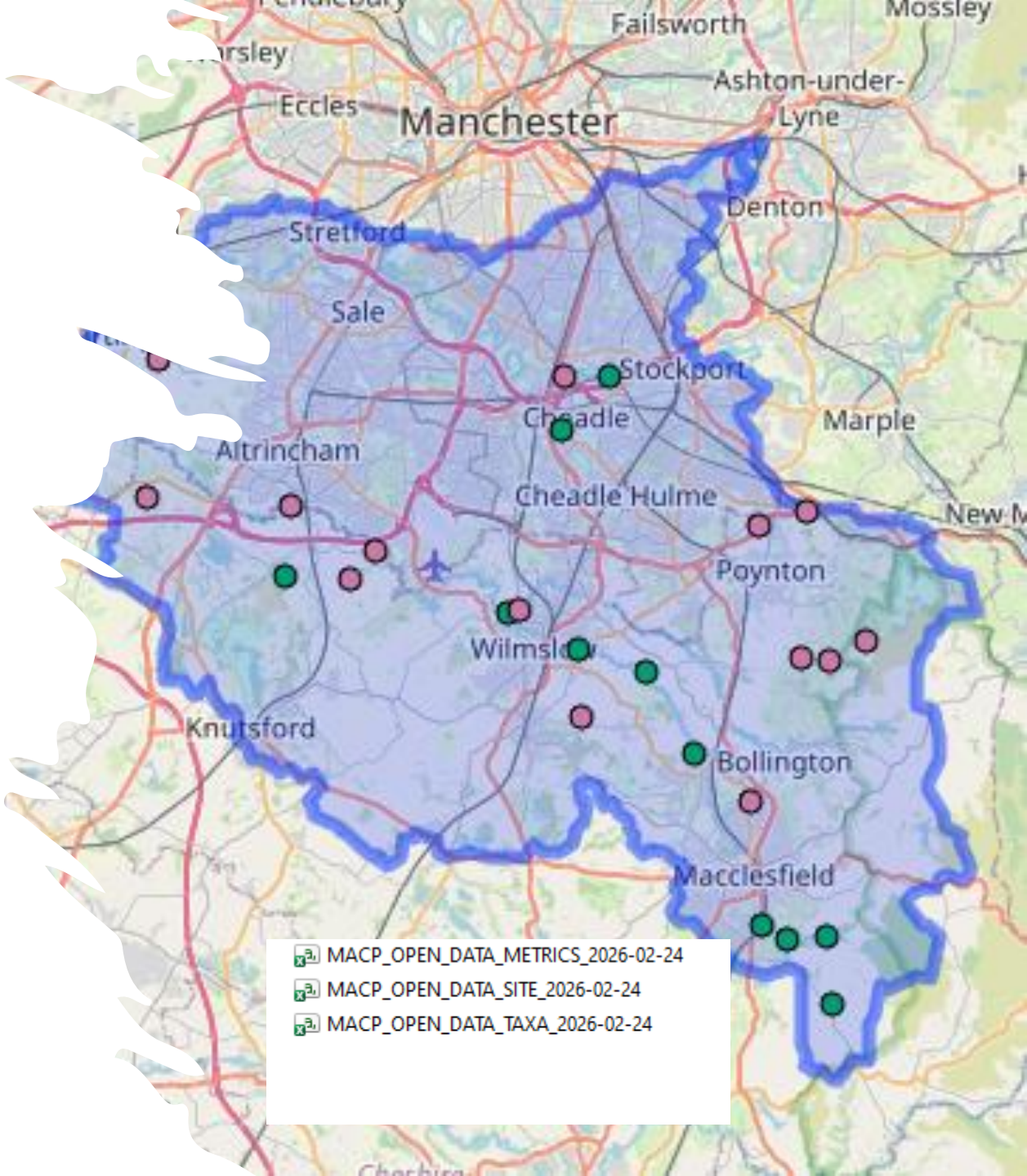
```
# Site dates are in SURVEY_DATE column (format: DD/MM/YYYY)
site_dates <- as.Date(site_data$SURVEY_DATE, format = "%d/%m/%Y")
site_data$YEAR <- as.numeric(format(site_dates, "%Y"))
```

Enough of me talking.. Let's work!

https://github.com/Leuctridae/Machine-Learning-in-Ecology-R-Workshop-

Complete the workshop before going to last slide!

Results & next steps

- Go through the results on Github, go through them.
- There are other data to add, can you do it? (macrophytes). Take your time, rerun the models.
- Go back to the data fish explorer page, extend the date range from 2020-2026 to 2000-2026 (same sites, re-run the script)
- Can you get that % Var explained higher than: 0.52%?
- Next steps.. Explore other methods!

https://environment.data.gov.uk/ecology/explorer/